Exercise 4 - SQL JOINS, UNION FILTERING & AGGREGATES

1. SQL JOINS

--- Query1 INNER JOIN

```
SELECT  E.EmployeeID,
        E.FirstName,
        E.LastName,
        E.Department,
        E.Salary,
        P.ProjectID,
        P.ProjectName,
        P.Budget,
        P.Status
FROM Employees AS E
INNER JOIN Projects AS P
ON E.EmployeeID = P.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Nilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

--- Query2: LEFT JOIN

```
SELECT  E.EmployeeID,
        E.FirstName,
        E.LastName,
        E.Department,
        E.Salary,
```

```
        P.ProjectID,
        P.ProjectName,
        P.Budget
        P.Status
FROM Employees AS E
LEFT JOIN Projects AS P
ON E.EmployeeID = P.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	~~HR~~ IT	70000	103	Cybersecurity Audit	75000	Pending ~~Ongoing~~
2	Alice	Smith	~~Finance~~ HR	~~75000~~ 60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	~~72000~~ 75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	~~65000~~ 72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

```
-- Query 3: Right Join
SELECT  P.ProjectID,
        P.ProjectName,
        P.Budget,
        P.Status,
        E.EmployeeID,
        E.FirstName,
        E.LastName,
        E.Department,
        E.Salary
FROM Employees AS E
RIGHT JOIN Project AS P
ON E.EmployeeID = P.EmployeeID;
```

ProjectID	ProjectName	Budget	Status	EmployeeID	FirstName	LastName	Department	Salary
101	AI Development	100000	Completed	1	John	Doe	IT	70000
102	Employee Training	50 000	Ongoing	2	Alice	Smith	HR	60000
103	cybersecurity Audit	75000	Pending	1	John	Doe	IT	70000
104	Financial Analysis	90000	Ongoing	3	Bob	Johnson	Finance	75000
105	Market Expansion	65000	Completed	5	Emma	Wilson	Sales	65000
106	Risk management	80000	Pending	6	Michael	Clark	Finance	80 000

--- Query 4 : Full Outer Join

```
SELECT E.EmployeeID,
       E.FirstName,
       E.LastName,
       E.Department,
       E.Salary,
       P.ProjectID,
       P.ProjectName,
       P.Budget,
       P.~~Salary~~ Status
FROM Employees AS E
FULL OUTER JOIN Projects AS P
ON E.EmployeeID = P.ProjectID ;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status ~~Salary~~
1	John	Doe	IT	70000	101	AI Development	100 000	Completed
1	John	Doe	IT	70000	103	cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	NULL 104/107	Financial Analysis	90 000	Ongoing
4	David	Brown	IT	72000	NULL ~~105~~	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80 000	106	Risk management	80000	Pending

3. Filtering Statements

--- Query 7

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
FROM   Employees
WHERE  Salary > 70000;
```

EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

-- Query 8

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
FROM   Employees
WHERE  Department IN ('IT', 'Finance');
```

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70 000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

--- Query 9

```
SELECT ProjectID,
       ProjectName,
       Budget,
       Status
FROM Projects
WHERE Status != 'Completed';
```

ProjectID	ProjectName	Budget	Status
102	Employee Training	50 000	Ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80 000	Pending

--- Query 10

```
SELECT ProjectID,
       ProjectName,
       Budget,
       Status
FROM Projects
WHERE budget > 70000 AND Status != 'Completed';
```

Project ID	ProjectName	Budget	Status
~~101~~	~~AI Development~~	~~100 000~~	
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80 000	Pending

--- Query 11

```
SELECT EmployeeID,
       FirstName,
       lastName,
       Department,
       Salary , City
FROM Employees
WHERE City IN('New York', 'Toronto')
ORDER BY Salary DESC;
```

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

--- Query 12

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary ~~Ass~~ ~~Mr~~ AS ~~Highest~~ Highest_paid
FROM Employees
ORDER BY Salary DESC;
LIMIT 3;
```

EmployeeID	FirstName	LastName	Department	Highest_paid
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

4. Aggregate Functions with GROUP BY & HAVING

```
SELECT Department,
       SUM(Salary) AS TotalSalary
FROM Employees
ORDER BY TotalSalary DESC;
```

Department	TotalSalary
Finance	155 000
IT	142 000
Sales	65 000
HR	60 000

```
--- Query 14
SELECT City,
       AVG(Salary) AS AverageSalary
FROM Employees
GROUP BY City
HAVING AVG(Salary) > 65 000;
```

City	AverageSalary
New York	75 000
Toronto	75 000
London	72 000

```
--- Query 15
SELECT Department,
       COUNT(Distinct EmployeeID) AS EmployeeCount
FROM Employees,
GROUP BY Department
HAVING COUNT(DISTINCT EmployeeID) > 1
```

Department	EmployeeCount
IT	2
Finance	2

--- Query 16

```
SELECT  Status,
        COUNT(ProjectID)  AS  ProjectCount
FROM Projects
GROUP BY (Status)
HAVING  COUNT(ProjectID) ≥ 2 ;
```

Status	ProjectCount
Completed	2
Ongoing	2
Pending	2

--- Query 17

```
SELECT E.EmployeeID,
       E.FirstName,
       E.LastName
       SUM(P.Budget)  AS  TotalProjectBudget
FROM  Employees  AS  E
INNER JOIN  Projects AS P      --- only want employees with projects
ON  E.EmployeeID = P.EmployeeID
GROUP BY E.EmployeeID,         --- list all columns in SELECT
         E.FirstName,
         E.LastName
HAVING SUM(Budget) > 150000 ;
```

EmployeeID	FirstName	LastName	TotalProjectBudget
1	John	Doe	175 000